- closed-loop mowing runs
- open-loop PRBS excitation
- randomized plant draws

Chapter 4

- trained offline on the exports
- rollout training objectives
- ablation-gated acceptance

Chapter 5

- PhysX mechanics at 60 Hz
- 64 parallel environments
- staged residual authority

Chapter 6

- same trajectories and metrics
- robustness vs adaptation split
- model- and control-level evidence

Chapter 7

Analytic where structure is reliable (PhysX rigid bodies), learned where modeling is the bottleneck (the hydraulic chain).